

Raw Playwright Output Divergence — same broken locator, different outputs

Scope: RAW Playwright artifacts only (`test-results.json` , `trace`, `stdout`, `stderr`).
LevelUp processing deliberately ignored. Goal: explain why the same broken locator (`#username` , which does not exist on SauceDemo — the real field is `#user-name` / `[data-test="username"]`) yields different Playwright outputs, and name the root cause to investigate next.

Test repo: `LevelUpAI_SauceDemo` . Broken locator lives in `pages/LoginPage.ts:6`
(`username = this.page.locator('#username')`), used at line 11 (`await this.username.fill(user)`), driven by `tests/verify-successful-login-with-valid-credentials.spec.ts` .

The three raw outcomes I reproduced (identical code, identical locator)

	Run A — natural action timeout	Run B — killed mid-action (SIGINT / process-group kill)	Run C — aborted before action timeout (global/budget timeout)
How produced	<code>npx playwright test <spec> (30s test timeout fires)</code>	start run, <code>kill -INT -<pgid></code> at 6s	<code>npx playwright test <spec> --global-timeout=8000</code>
test-results.json written?	✅ yes	❌ no file at all	✅ yes
Failing test result status	<code>timedOut</code>	(none)	<code>skipped / interrupted</code>
Per-action error present?	✅ <code>errors[1] = locator.fill ... waiting for locator('#username')</code>	—	❌ none
<code>errors[].location</code>	✅ <code>pages/LoginPage.ts:11</code>	—	❌ none (only top-level <code>errors[]</code>)
Only error available	located action error	—	generic "Timed out waiting 8s for the test suite to run" (no location, no locator)
trace.zip produced?	✅ yes	❌ no	❌ no
Locator recoverable from raw output?	✅ <code>#username</code>	❌ nothing to parse	❌ nothing to parse

Run A raw (the GOOD shape — matches the on-disk captured CI run)

```
status: timedOut
errors[0].message: "Test timeout of 30000ms exceeded." location: None
errors[1].message: "Error: locator.fill: ... waiting for locator('#username')"
errors[1].location: { file: pages/LoginPage.ts, line: 11, column: 25 }
```

stdout literally prints `- waiting for locator('#username')`. trace.zip attached.

This is exactly the shape of the failing run already captured on disk

(`test-results/.../error-context.md` shows the same located `waiting for locator('#username')`).

Run B raw (kill mid-action)

```
(no test-results.json written – Playwright never finished the reporter)
stdout stops after "[1/1] ... Verify successful login"
```

Run C raw (the BAD shape — reproduces the hallucination input)

```
failing test result -> status: skipped, error: None, errors: [], errorLocation: None
top-level errors: [
  "Timed out waiting 8s for the test suite to run",
  "Timed out waiting 8s for the teardown for test suite to run"
]
```

No per-test error, **no location anywhere**, no trace. The only signal is a generic suite-level timeout string that never mentions `#username`.

Why the same broken locator produces different outputs

A non-existent locator in Playwright is **not** an immediate error — `locator.fill()` waits. The detailed, located error (`waiting for locator('#username') + location: LoginPage.ts:11`) is only emitted **when the per-action / per-test timeout fires naturally**. Everything downstream that can recover `#username` depends on that one event happening.

So the divergence is entirely a function of **how the run ends**:

1. **Allowed to reach the 30s action/test timeout (Run A):** Playwright produces the rich located error + call log + trace. `#username` is in the raw output. This is the original CI execution's shape (proven by the on-disk `error-context.md`).
2. **Terminated before the action timeout fires (Runs B & C):** the wait is cut short, so Playwright never emits the located action error. You get either **no results file** (hard kill, Run B) or a **generic suite-level timeout with no location and no locator** (graceful global/budget abort, Run C). `#username` is absent from the raw output. This is the shape that starves extraction and lets the AI hallucinate.

Same locator. The difference is timing of termination, not the locator and not the healing pipeline.

Root cause to investigate next (raw-execution layer, NOT the healing pipeline)

The original CI failure is Run A (located, has `#username`). The hallucination came from a **re-execution that ended like Run C/B**. The execution layer is what changes the ending:

- `ExecutionEngine.runAsync` runs the rerun under a **budget-derived timeout** (`rerunTimeoutMs()` computed from remaining `HEALING_JOB_BUDGET_MS` / `HEALING_PER_TEST_BUDGET_MS`) and **kills the process group** (`process.kill(-pid)`) on timeout. When the remaining budget is smaller than the test's own 30s action timeout, the rerun is killed **before** Playwright emits the located error → Run B (no file) or Run C (generic, no-location) shape → `failed_locator = null` → AI hallucination.
- A grep-only / different rerun target (whole-suite) can also change which timeout fires first and whether a per-action error is ever produced.

Investigate: the relationship between `rerunTimeoutMs()` (budget-derived) and the test repo's own action/test timeout (30s here). Whenever rerun timeout $\leq$ action timeout, the rerun cannot produce a located locator error — guaranteeing the no-location / no-file shape. Confirm with the failing job's actual rerun timeout and exit signal.

To close the last 1%

The single field that distinguishes the two shapes in the failing job's raw

`test-results.json`:

```
.suites[].specs[].tests[].results[].errors[].location # present = Run A ; absent
(only top-level errors[]) = Run C
```

Plus the rerun's exit signal (clean exit vs SIGKILL/SIGTERM) and the effective rerun timeout vs the repo's 30s action timeout.

Reproduction commands (exact)

```
cd LevelUpAI_SauceDemo
# Run A – GOOD shape (located #username, trace, status timedOut)
npx playwright test tests/verify-successful-login-with-valid-credentials.spec.ts

# Run B – killed mid-action (no results file)
setsid npx playwright test tests/verify-successful-login-with-valid-credentials.spec.ts & \
  PGID=$!; sleep 6; kill -INT -$PGID

# Run C – BAD shape (generic suite timeout, no location, no locator, no trace)
npx playwright test tests/verify-successful-login-with-valid-credentials.spec.ts --
global-timeout=8000
```